



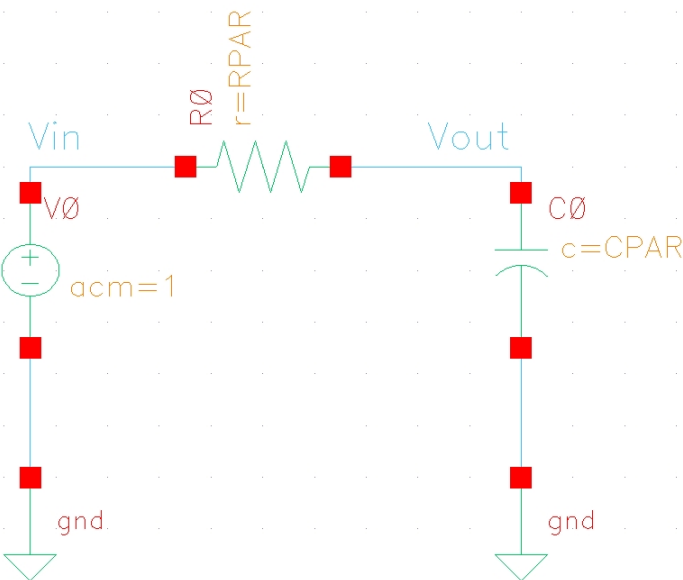
ITI Cmos Analog IC Design
Lab 10
Noise Simulation

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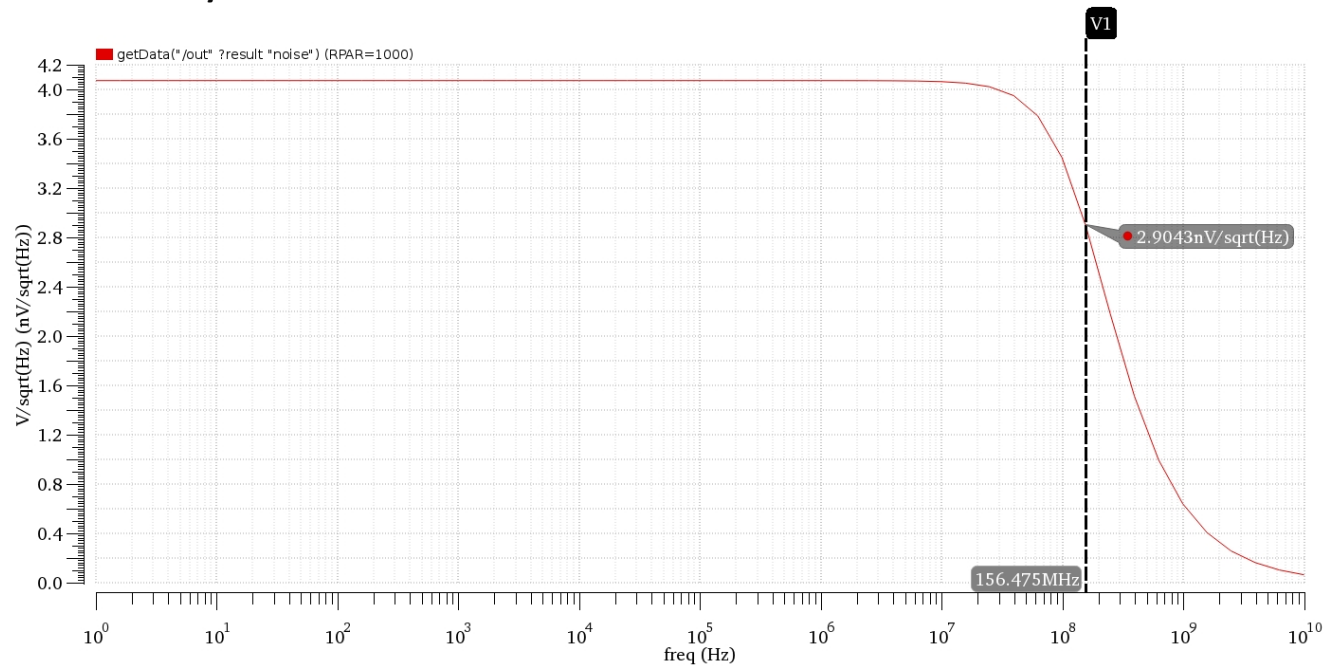
Section 7 online

Part 1 — LPF AC Noise Analysis

1.1 Circuit schematic



1.2 Noise Analysis



my_labs:noise_lab09:1	ymax(getData("/out" ?result "...	4.071n
my_labs:noise_lab09:1	RMS_Noise	65.16u
my_labs:noise_lab09:1	BW	158.8M

Hand Analysis:

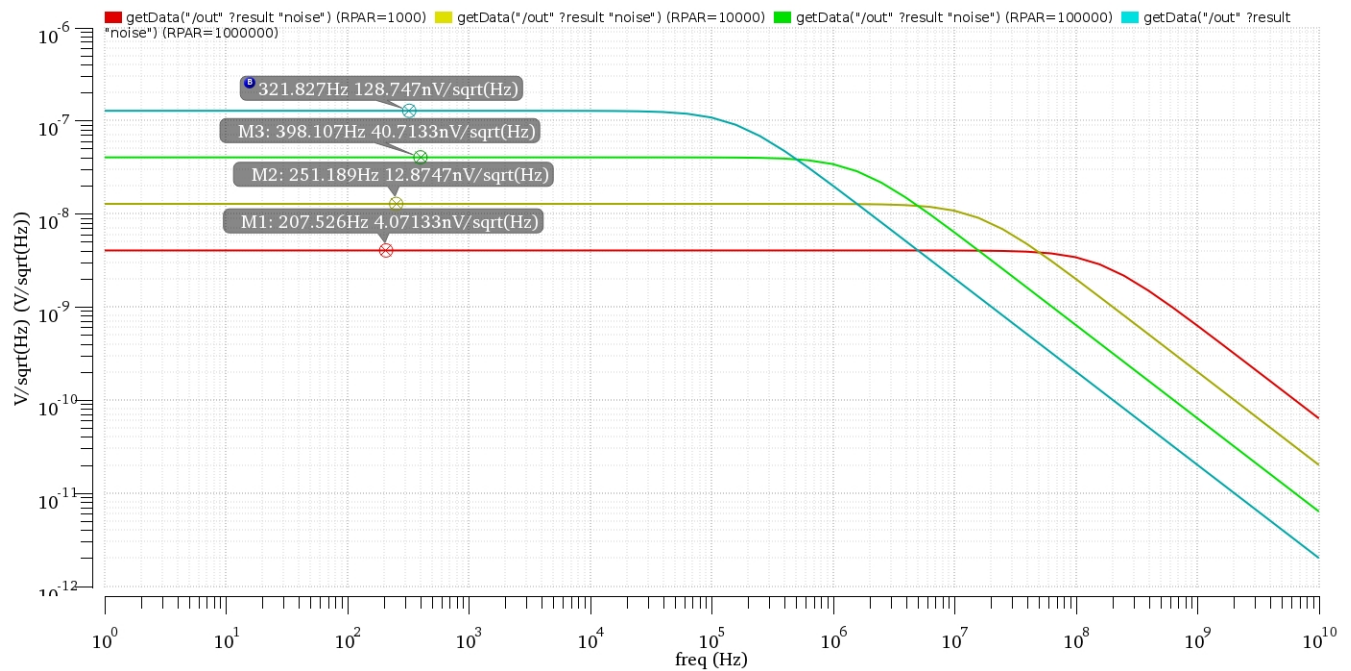
Voltage noise density = $4 \text{ KTR} = 4 \text{ nV}/\sqrt{\text{Hz}}$

RMS Noise = $kT/C = 64 \text{ uVrms}$

BW = $1 / 2 * \text{Pi} * R * C = 159.15 \text{ MHz}$

	Noise Density (V/ $\sqrt{\text{Hz}}$)		Bandwidth (Hz)		RMS Noise (V)	
	Simulation	Hand analysis	Simulation	Hand analysis	Simulation	Hand analysis
Noise	4.071 n	4 n	158.8M	159.1M	65.15u	64 u

1.5 Parametric Sweep — RMS Noise for Different R Values



Parameters: RPAR=1k			
1	my_labs:noise_lab09:1	RMS_noise	65.16u
1	my_labs:noise_lab09:1	density	4.071n
1	my_labs:noise_lab09:1	BW	158.8M
Parameters: RPAR=10k			
2	my_labs:noise_lab09:1	RMS_noise	65.48u
2	my_labs:noise_lab09:1	density	12.87n
2	my_labs:noise_lab09:1	BW	15.88M
Parameters: RPAR=100k			
3	my_labs:noise_lab09:1	RMS_noise	65.51u
3	my_labs:noise_lab09:1	density	40.71n
3	my_labs:noise_lab09:1	BW	1.588M
Parameters: RPAR=1M			
4	my_labs:noise_lab09:1	RMS_noise	65.51u
4	my_labs:noise_lab09:1	density	128.7n
4	my_labs:noise_lab09:1	BW	158.8k

R	Noise Density (V/√Hz)	Bandwidth (Hz)	RMS Noise (V)
1 kΩ	4.071 n	158.8 MHz	65.16 u
10 kΩ	12.87 n	15.88 MHz	65.16 u
100 kΩ	40.71 n	1.588 MHz	65.16 u
1000 kΩ	128.7n	158.8 kHz	65.16 u

As we can see

Noise density is depend on value of resistor = $4KTR$.

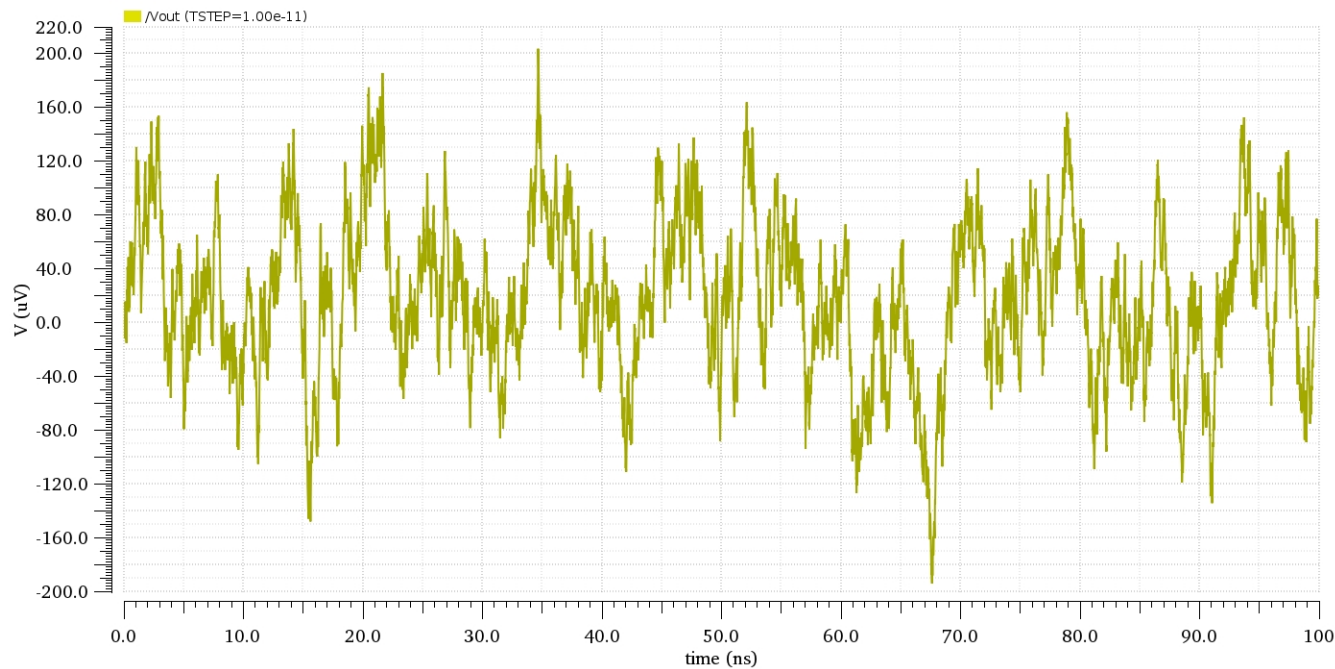
Rms noise Value is not depened on Resistor and only depend only on $C = KT/C$.

The reason for that is while noise Density increase with resitance , bandwidth of transfer function only decrease by the same value so rms still constant

$$Bw = 1 / 2 * \pi * R * C$$

Part 2 — LPF Transient Noise Analysis

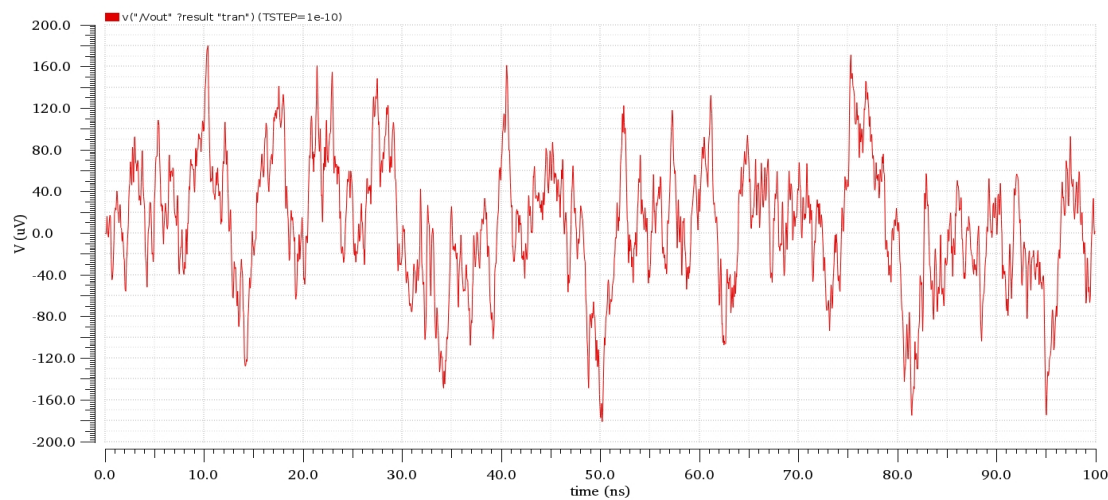
2.1 Transient Noise (TSTEP = TAU/100)



my_labs:noise_lab09:1	rms(v("/Vout" ?result "tran"))	60.88u
my_labs:noise_lab09:1	ymax(v("/Vout" ?result "tran"))	203.8u
my_labs:noise_lab09:1	ymin(v("/Vout" ?result "tran"))	-194.2u

Quantity	Part 1 (AC Noise)	Part 2 (Transient Noise, TSTEP = TAU/100)
RMS noise (V)	65.16 u	60.88 u

2.2 TSTEP to TAU/10



my_labs:noise_lab09:1	rms(v("/Vout" ?result "tran"))	61.59u
my_labs:noise_lab09:1	ymin(v("/Vout" ?result "tran"))	180.2u
my_labs:noise_lab09:1	ymin(v("/Vout" ?result "tran"))	-181u

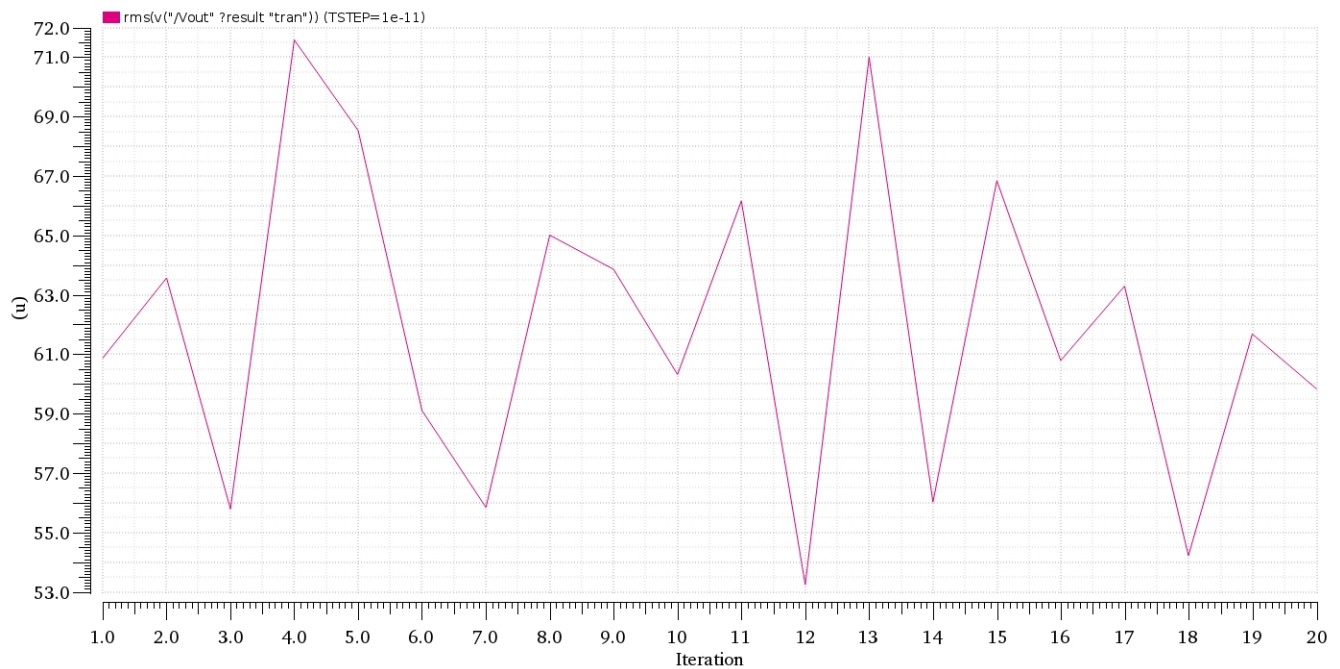
Rms noise = 61.59u

value increase

after step is increase the value must be less accurate than one with lower step

2.6 20-Run Transient-Noise Iteration Analysis

RMS Noise vs Iteration

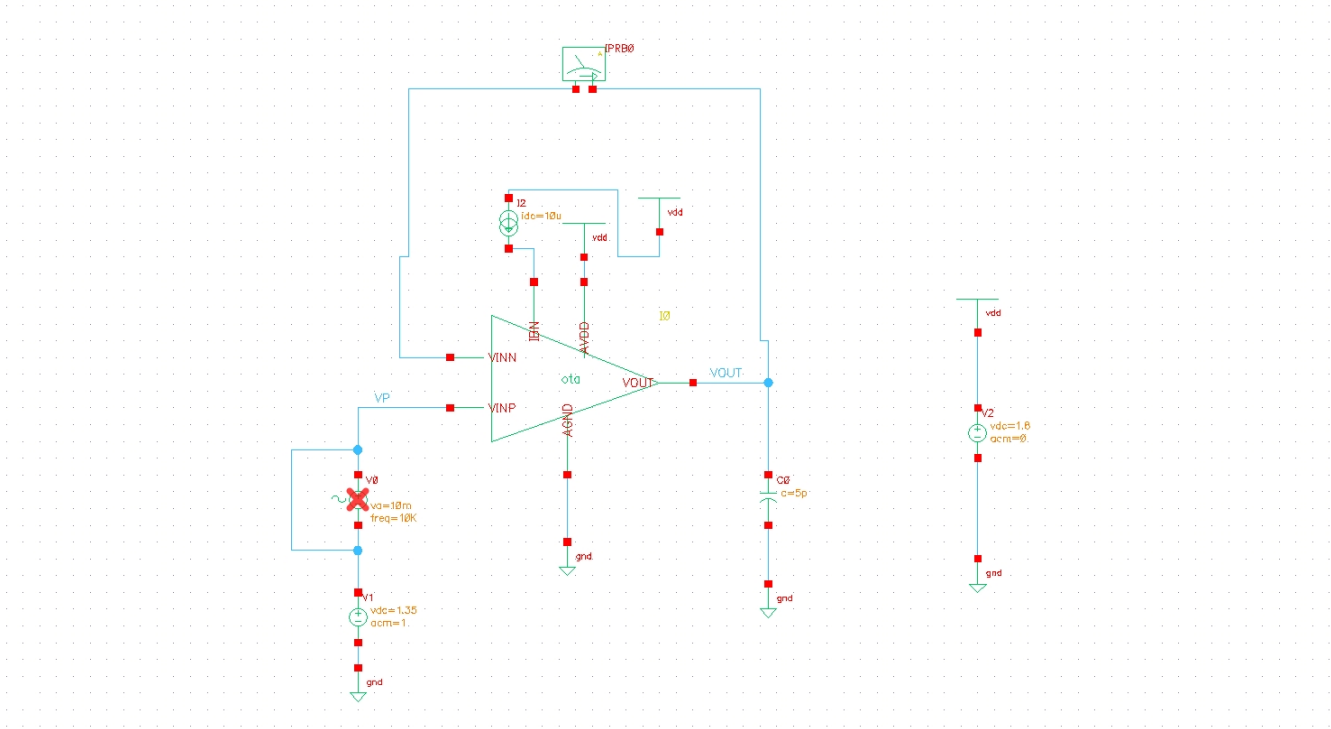


average RMS	61.97u
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Quantity	Part 1 (AC Noise)	Part 2 — Single Run	Part 2 — Average of 20 Runs
RMS noise (V)	65.16 u	60.88 u	61.97 u

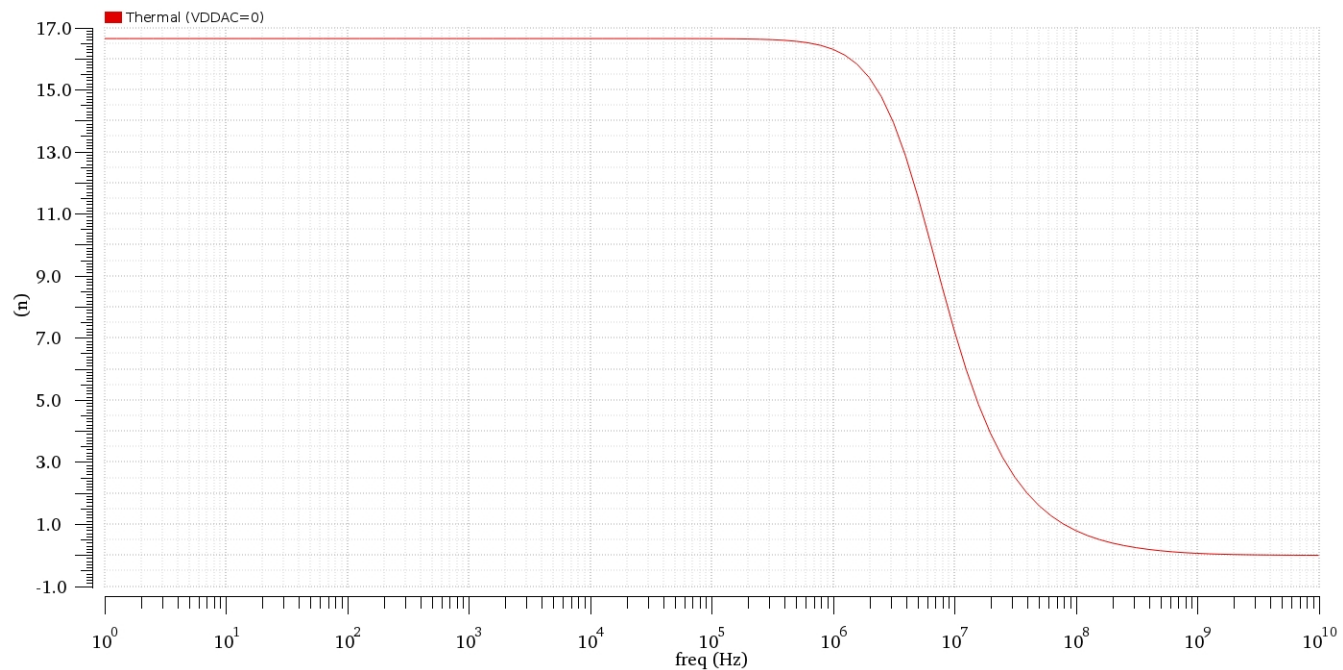
Part 3 — 5T OTA AC Noise Analysis

3.1 5T OTA Schematic

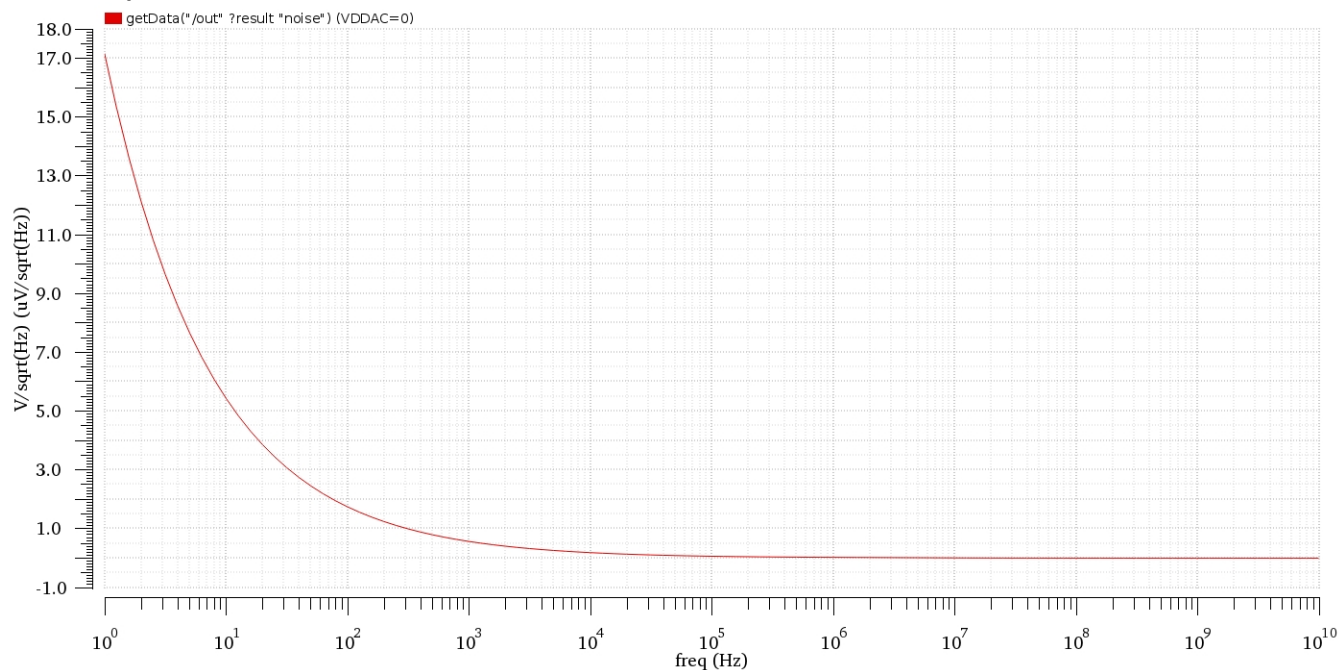


3.2 Output Noise vs Frequency

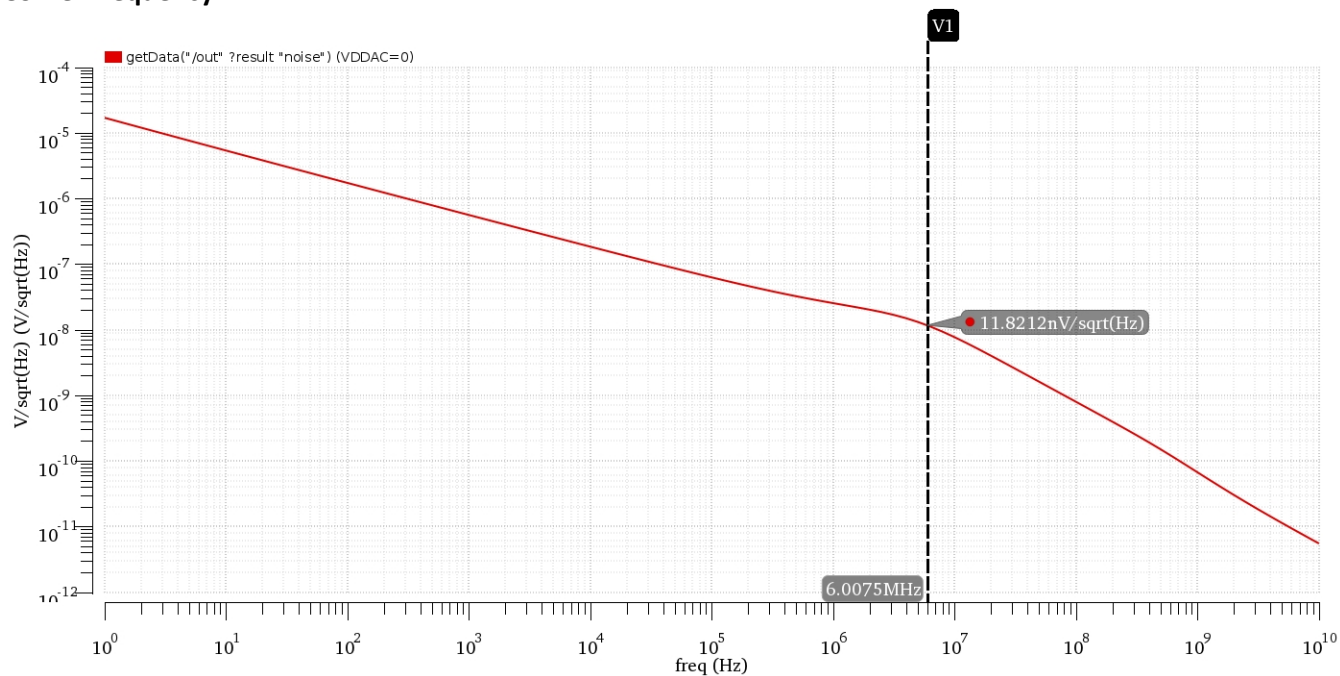
Thermal Noise



Total output Noise



Corner frequency



my_labs:ota_tb:1	rmsNoise(1 1e+10)	86.34u
my_labs:ota_tb:1	max thermal density	16.66n
my_labs:ota_tb:1	rms Thermal	46.11u
my_labs:ota_tb:1	bandwidth(sqrt((pv("IO.M0.m...	4.842M

Hand Analysis

For a 5T OTA, the input-referred thermal noise density is approximately:

$$\text{Noise thermal denesity} = \sqrt{S_v} \approx \sqrt{8kT/gm_1 [1 + gm_3/gm_1 \cdot \dots]} = 8(4.14 \cdot 10^{-21})/159\mu [1+0.88] \\ = 16.14\text{n} \quad (\text{V}/\sqrt{\text{Hz}})$$

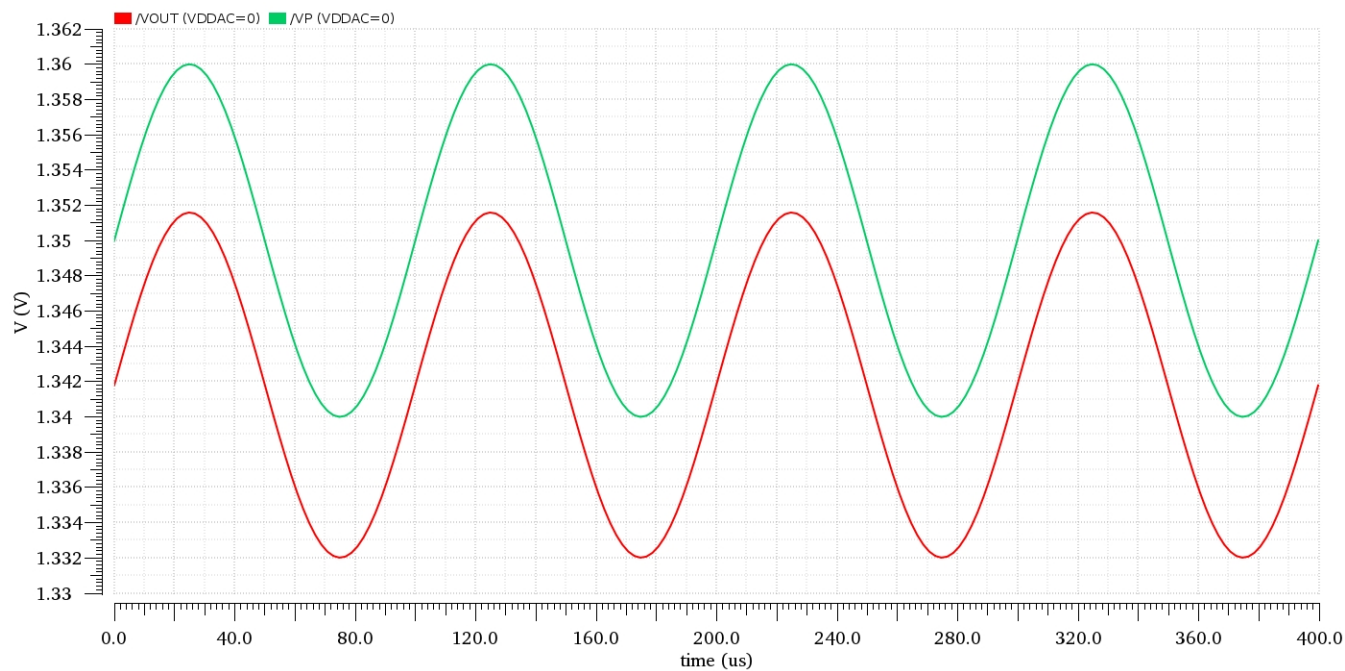
$$Wp_CL \text{ (Bandwidth)} = Wp_OL * (1+Aol) = GBW = 5 \text{ MHz}$$

$$\text{Rms Thermal} = \sqrt{S_v * BN} = (16.14\text{n})^2 * \pi/2 * 5\text{M}) = 45.52\mu\text{Vrms}$$

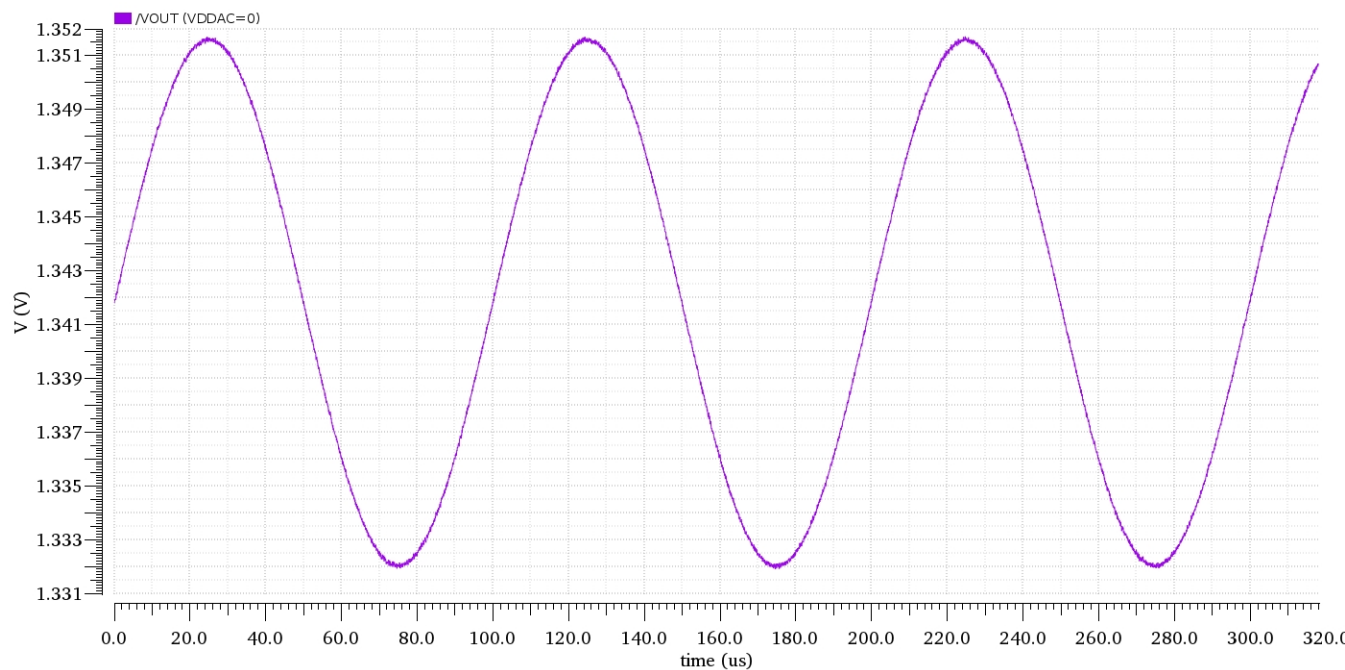
Quantity	Hand Analysis	Simulation
Noise density (V/√Hz)	16.14n	16.66n
Bandwidth (Hz)	5 Mhz	4.8 Mhz
Rms Value (Thermal)	45.52u	46.11u

Part 4 — 5T OTA Transient Noise Analysis

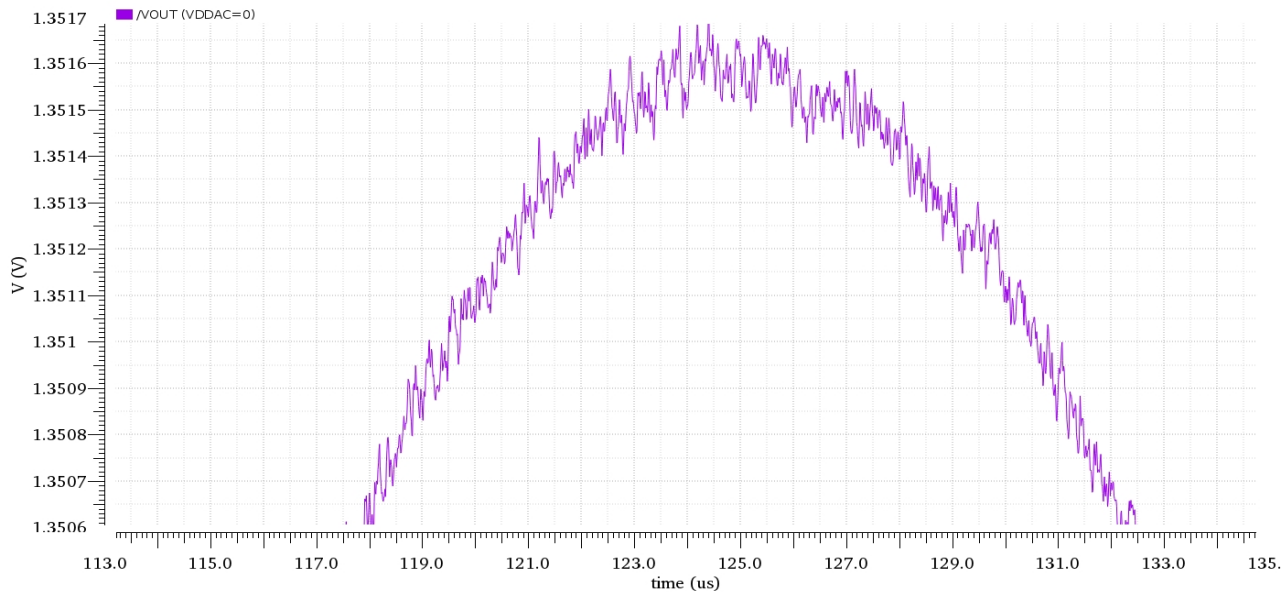
4.1 Input/Output Waveform Comparison



4.2 Transient Noise

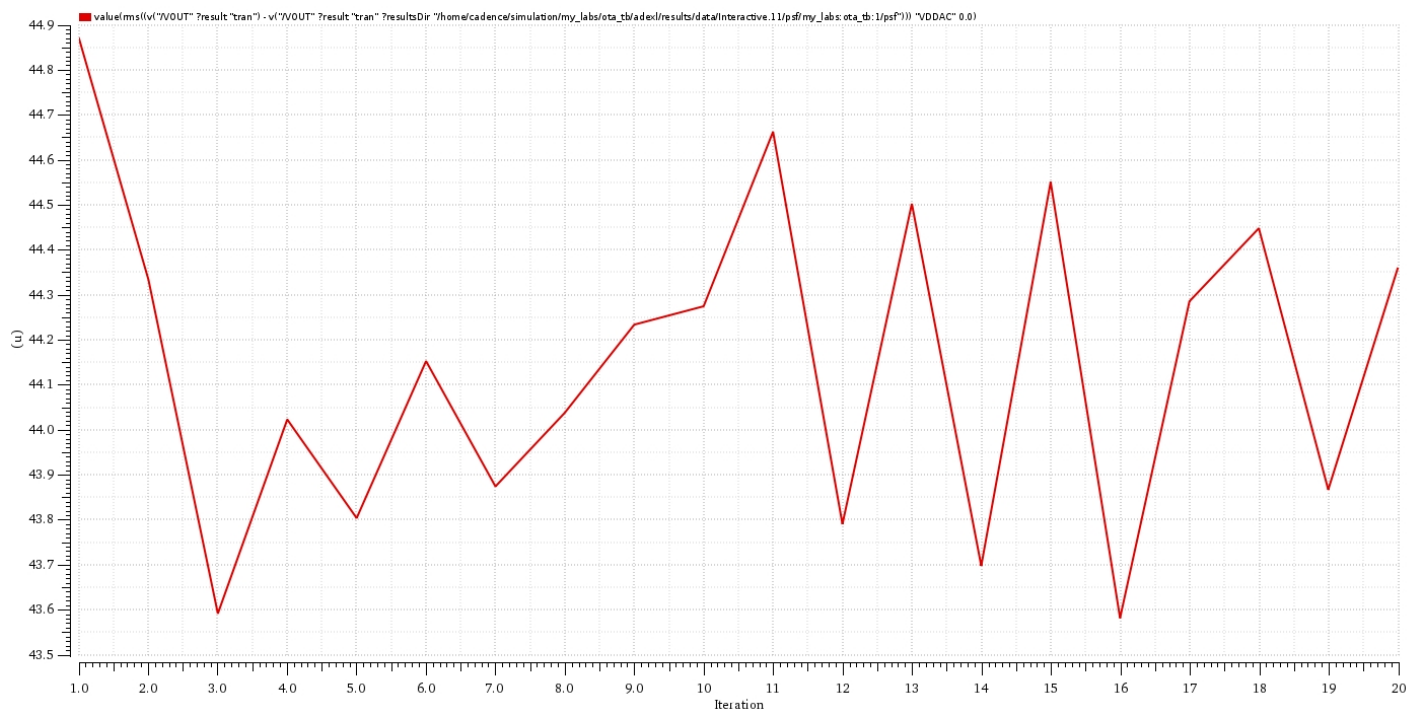


Zoom in picture



4.3 20 Transient-Noise Runs

RMS Noise vs Iteration



average_rms

44.16u

Comparison with Part 3

Quantity	Part 3 (AC Noise)	Part 4 (Average of 20 Transient Runs)
RMS noise (V)	46.11u	44.16u
